

DBMS SKILL 5

```
1 from fastapi import FastAPI
2 from sqlalchemy import create_engine, text
3 from sqlalchemy.engine import URL
4 from pydantic import BaseModel
5
6 app = FastAPI()
7
8 DATABASE_URL = URL.create(
9     drivename="mysql+pymysql",
10    username="root",
11    password="nishanth@123",
12    host="localhost",
13    port=3306,
14    database="university"
15 )
```

blems Output Debug Console Terminal Ports

```
INFO: Started server process [12576]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:55076 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:55076 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:55076 - "GET /students HTTP/1.1" 200 OK
INFO: 127.0.0.1:55089 - "POST /students HTTP/1.1" 200 OK
```

```
17 engine = create_engine(DATABASE_URL)
18
19
20 class Student(BaseModel):
21     id: int
22     name: str
23     department: str
24
25
26 @app.get("/students")
27 def get_students():
28     with engine.connect() as connection:
29         result = connection.execute(
30             text("SELECT * FROM students")
31         )
```

```
36 @app.post("/students")
37 def create_student(student: Student):
38     with engine.begin() as connection:
39         connection.execute(
40             text("""
41                 INSERT INTO students (id, name, department
42                 VALUES (:id, :name, :department)
43             """),
44             {
45                 "id": student.id,
46                 "name": student.name,
```

```
46                 "name": student.name,
47                 "department": student.department
48             }
49         )
50
51     return {
52         "message": "Student created",
53         "student": student
54     }
```

default

GET

/students

Get Students

Parameters

No parameters

Cancel

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students
```

Server response

Code	Details
200	Response body [,

200

Response body

```
{
  {
    "id": 101,
    "name": "Ravi",
    "department": "CSE"
  },
  {
    "id": 102,
    "name": "Priya",
    "department": "ECE"
  }
}
```

Response headers

```
content-length: 90
content-type: application/json
date: Wed, 09 Sep 2026 09:39:17 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

POST

/students

Create Student

Parameters

No parameters

Request body

required

Edit Value | Schema

```
{
  "id": 103,
  "name": "Arun",
  "department": "CSE"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 103,
    "name": "Arun",
    "department": "CSE"
  }'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code	Details
200	Response body <pre>{ "message": "Student created", "student": { "id": 103, "name": "Arun", "department": "CSE" } }</pre> Download

Responses

Code	Description
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string", "input": "string", "ctx": {} }] }</pre>

PUT

/students/{student_id} Update Student

Parameters

Name	Description
student_id * required	103
integer	
(path)	

Request body required

Edit Value | Schema

```
{
  "id": 0,
  "name": "string",
  "department": "string"
}
```

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/103' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 0,
    "name": "string",
    "department": "string"
  }'
```

Request URL

http://127.0.0.1:8000/students/103

Server response

Code	Details
200	Response body <pre>{ "message": "Student updated", "id": 103, "name": "string", "department": "string" }</pre> Response headers <pre>content-length: 76 content-type: application/json date: Fri, 11 Sep 2026 07:05:42 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response

Name	Description
student_id * required integer (path)	103
ExecuteClear	

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/students/103' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students/103
```

Server response

Code	Details
200	Response body <pre>{ "message": "Student deleted", "id": 103 }</pre>

Code	Details
200	Response body <pre>{ "message": "Student deleted", "id": 103 }</pre> Response headers <pre>content-length: 38 content-type: application/json date: Fri, 11 Sep 2026 07:06:54 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response Media type application/json Controls Accept header.